

— GROUP 11 • BLUE OCEAN PITCH

HYPHA

Packaging that grows to fit.

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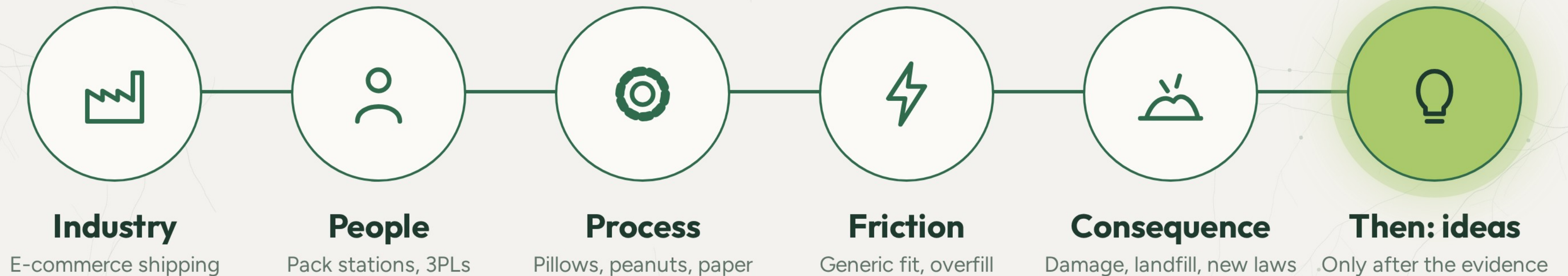


SET • READY TO SHIP

HYPHA PASTE

— BEFORE WE BUILT ANYTHING

We stayed in the problem first.



✓ Evidence before ideas

✓ Map the whole space

✓ Ask where it breaks

The problem space of packaging

Packaging



Materials

plastic foam paper

mycelium

- **Less than 10%** of plastic packaging is actually recycled, globally and in the US.
- Flexible films, plastic bags and rigid containers make up **about 40%** of global plastic waste.
- The US generates **40 million tons** of plastic waste a year; only **about 5%** gets recycled.



Cost

material labor shipping

size

- Recycled and compostable materials cost more than standard plastic or paper: **+28%** for recycled plastic, **+20%** for paper, depending on volume.
- Labor depends on where it's made: US-made costs **125% more**, but arrives **44% faster**.



Waste

landfill recycling carbon

- Packaging waste is projected to hit **434.5 million** metric tons.
- Packaging is **40%** of our plastic waste.
- Because of sorting issues, most packaging ends up in landfill.



Manufacturing

molds tooling inventory

- One multi-cavity injection mold for a custom bottle cap costs **\$50K to \$200K+** and takes **12 to 24 weeks** to make.
- Storing millions of empty bottles or boxes takes massive warehouse space.
- Switching a line from a 12oz to a 24oz bottle means swapping **dozens** of handling parts.



Customer

unboxing protection

experience branding

- A **5-experiment** Journal of Consumer Psychology study: watching a box being opened increased enjoyment.
- A 2025 study of **1,193** consumers: minimal white design, typography and texture felt more luxurious.
- **73%** say premium packaging means a better brand.
- **53%** with a good packaging experience returned to the brand.
- Protection is why shoppers prefer boxes over mailer bags (2025 e-commerce survey).

— THE PROBLEM

Today, the empty space gets **stuffed**.

Plastic air pillows

Foam peanuts

Crumpled paper

- ❌ **Generic fit**
Products shift and break in transit
- ❌ **Rarely recycled**
Most of it ends up in landfill
- ❌ **Under new laws**
EPR and polystyrene bans target it



Packaging is used once, then **buried**.



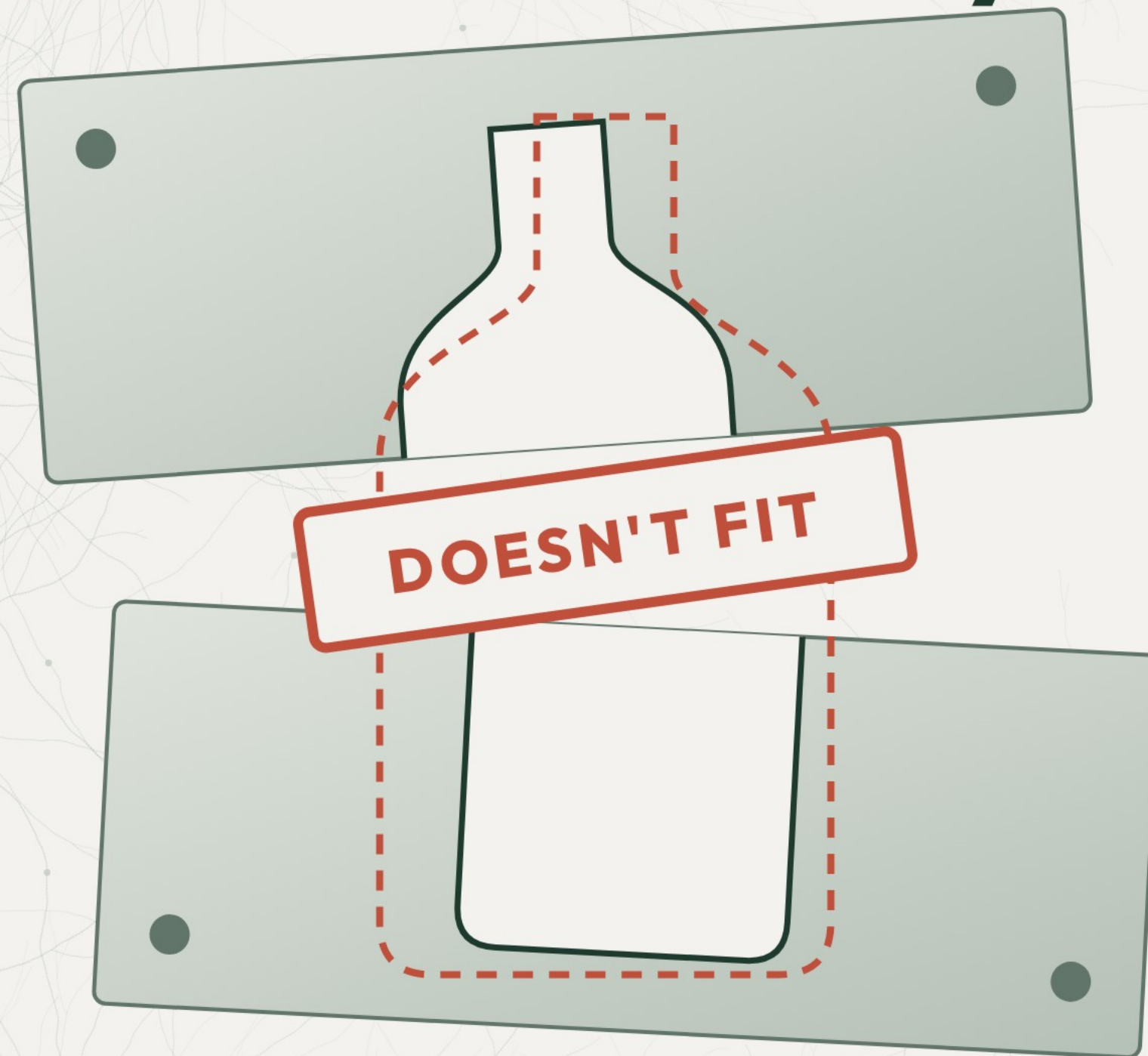
 1 box = 1% of US plastic waste

<10%

of that packaging is recycled.
The rest is buried or burned.

— THE OTHER OPTION • CUSTOM INSERTS

A custom fit today means a new mold.



ONE CUSTOM MOLD

\$200K+

starting around \$50K

BEFORE THE FIRST PART SHIPS

24 weeks



×2 Product changes? Pay it all again.

THE GREEN
PREMIUM

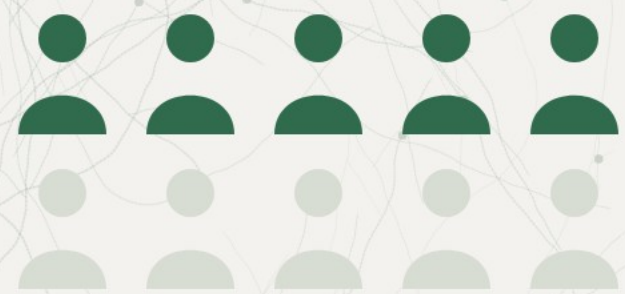
+28%
recycled plastic

+20%
recycled paper

Green premiums compare recycled with standard materials. Source: [10] Packaging Technology Today, "The Rise of Sustainable Packaging in the U.S."

The box is the brand's first handshake.

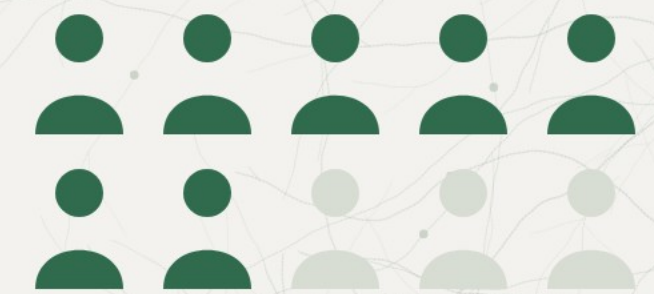
53%



came back after a great unboxing



73%

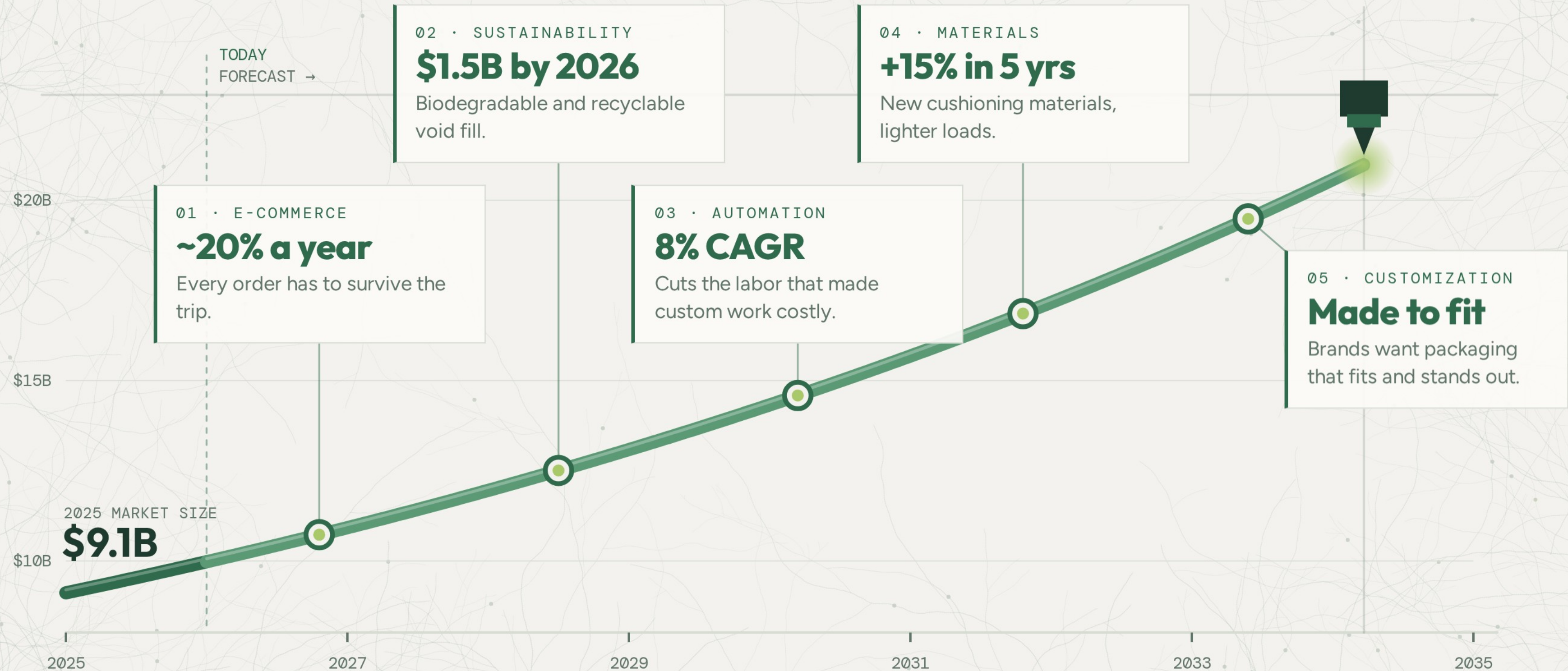


judge the brand by its packaging



It has to arrive intact. Protection is why shoppers pick boxes over mailers.

A \$9.1B market, compounding.



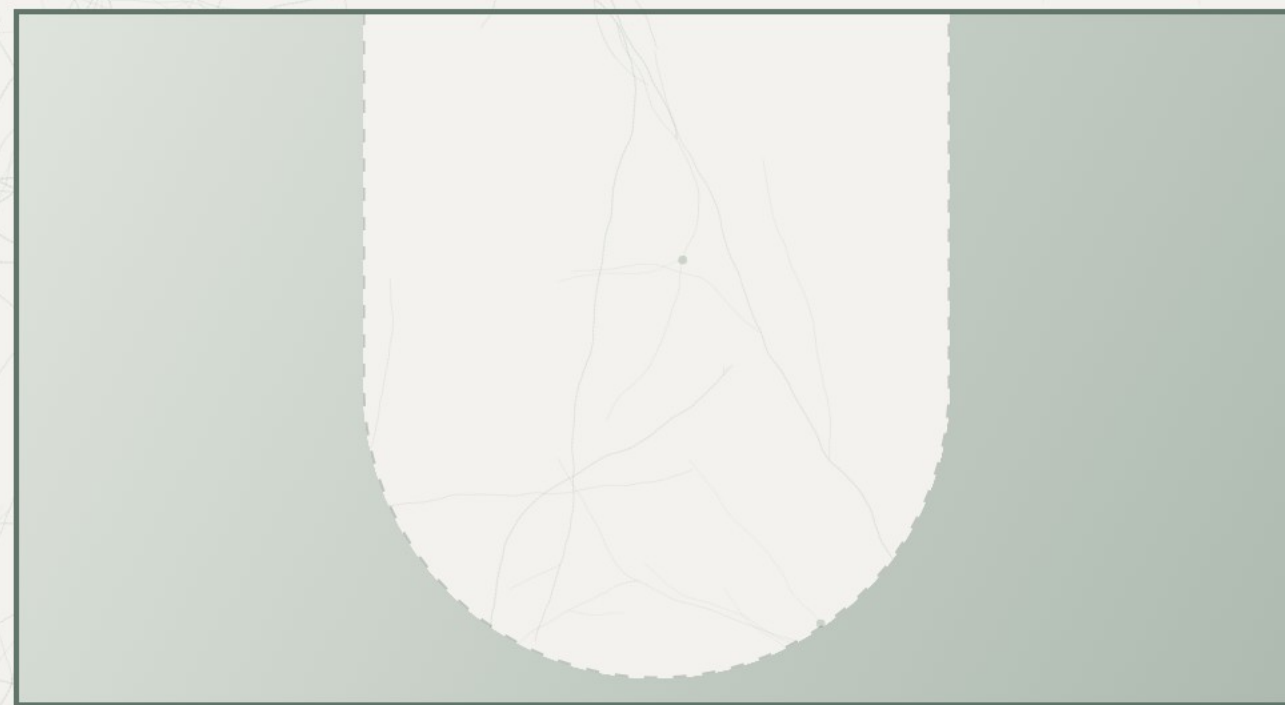
Market size: [1] Future Market Insights, US & Canada Protective Packaging Market (2025, 2035 forecast, 5.1% CAGR as reported). Drivers: [4] Market Research Future, Void Fill Packaging System Market. Curve between the two endpoints is interpolated.

— THE EXPONENTIAL DRIVER WE'RE BETTING ON

Put material only where it's needed.

Cut it out

TRADITIONAL



SCRAP

40% of the block becomes scrap

Build it up

ADDITIVE



SCRAP

0% scrap, and no mold

Illustration: the scrap share is measured from this drawing, not an industry figure.

— HOW IT WORKS

CONCEPT RENDER

Scan. Print. Expand. Return to soil.



Scan

A camera maps the box and product in 3D



Print

A nozzle traces only the empty space



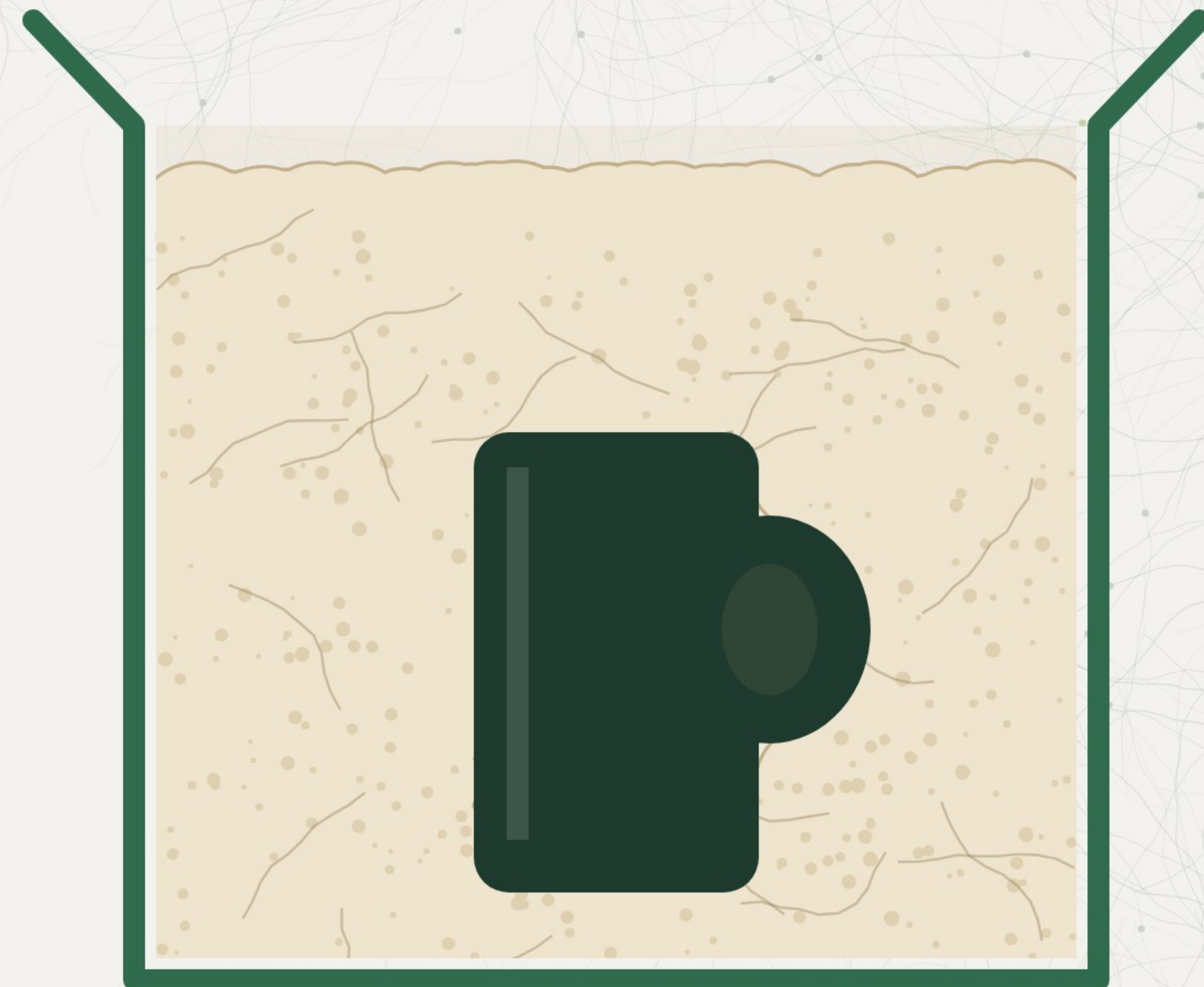
Expand

The paste foams and locks the product in



Return to soil

Composts at home in weeks



EXPANDING

HYPHA PASTE

— THE MATERIAL

Mushroom roots + farm waste.



Mycelium
the root network of
mushrooms

+



Hemp hurd
agricultural waste fiber

+

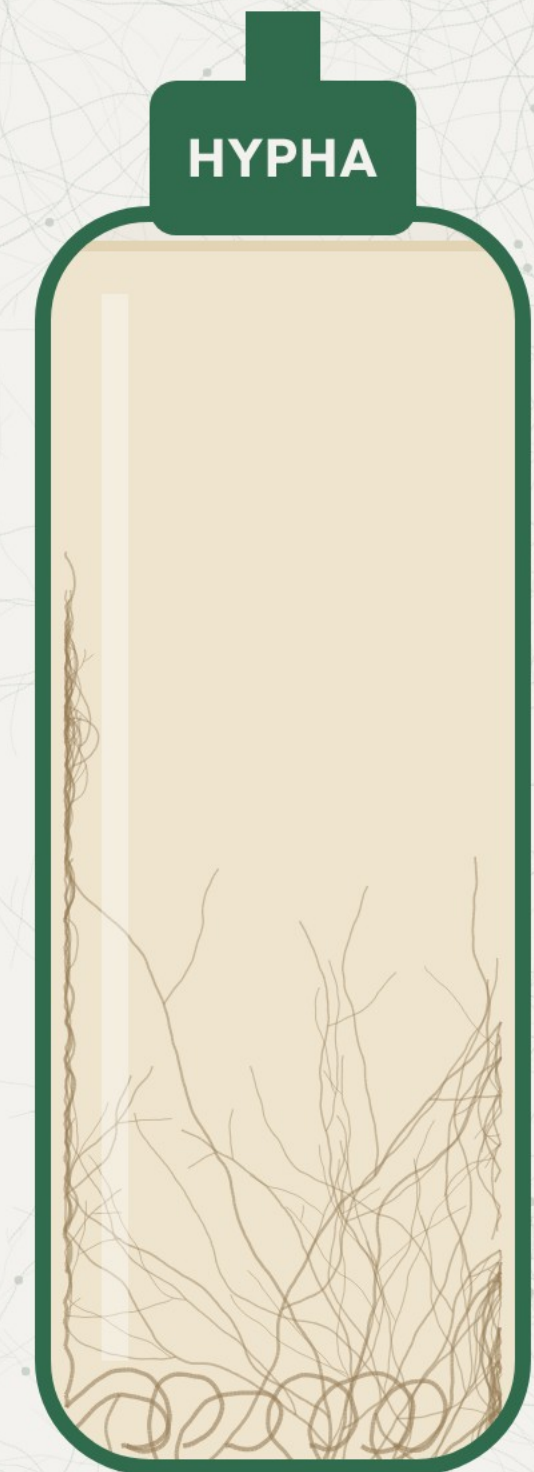


Corn husk
agricultural waste fiber

 Home-compostable

 Abundant & renewable

 Refillable cartridges



REFILLABLE CARTRIDGE

— WHO IT'S FOR

Built for the pack line.



Fulfillment centers



3PLs



DTC brands

Shipping fragile or oddly shaped products: electronics · cosmetics · home goods



Custom fit, every box

Fewer damaged shipments



Only what's needed

No overfill



Plastic-free

Home-compostable



Drop-in station

Sits at existing pack lines



A brand moment

Looks engineered, not stuffed

A real brand would switch, up to 2× the cost.

WHAT THEY USE TODAY

~\$1 per order

Boxes, poly mailers and poly sleeves, bought in bulk

"Roughly for all shipments, it's about a dollar per order cause we get all the packaging in bulk."

DAMAGE AND UNBOXING

Very important

Damage is rare, but the unboxing is part of the brand

"We try to make our packaging feel as personal as possible ... so the unboxing experience is very important to our brand."

WOULD THEY SWITCH?

<2× current cost

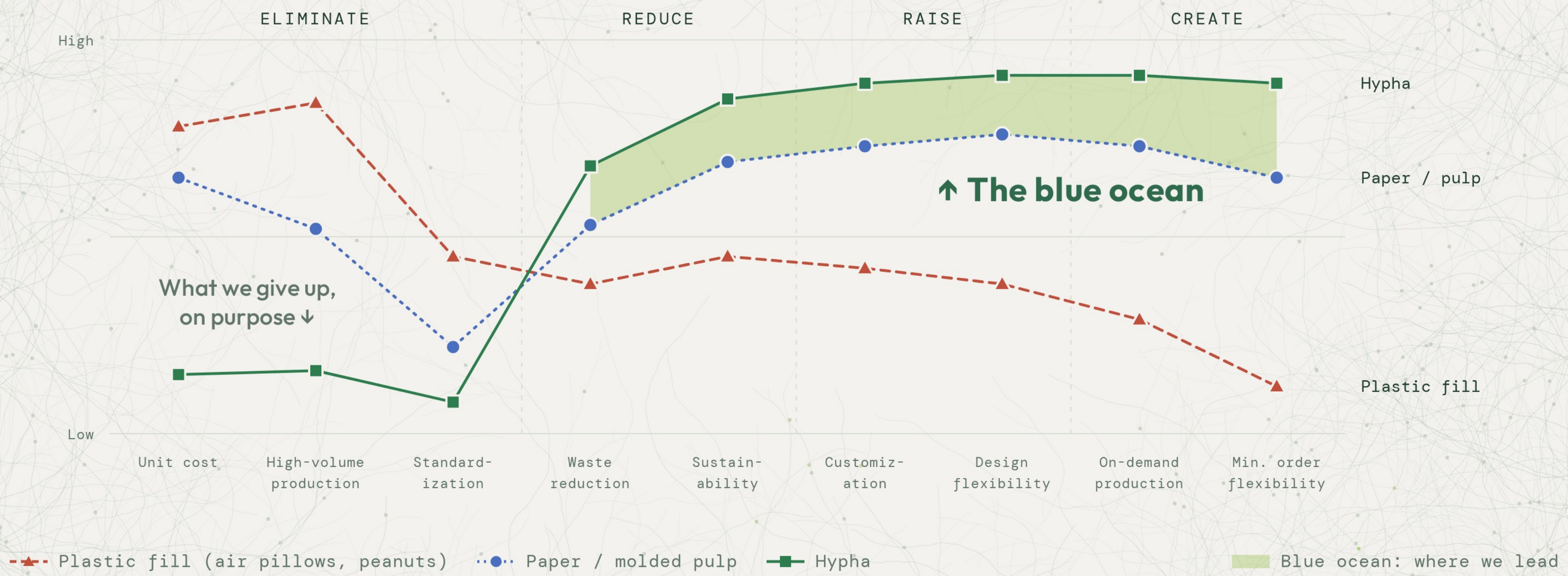
Yes, they'd trial it and possibly switch fully

"If it surpasses double our current shipping costs, that's when it stops making sense profit-wise."



Takeaway: Hypha has to land under ~\$2 per order, and the unboxing story sells it as much as sustainability does.

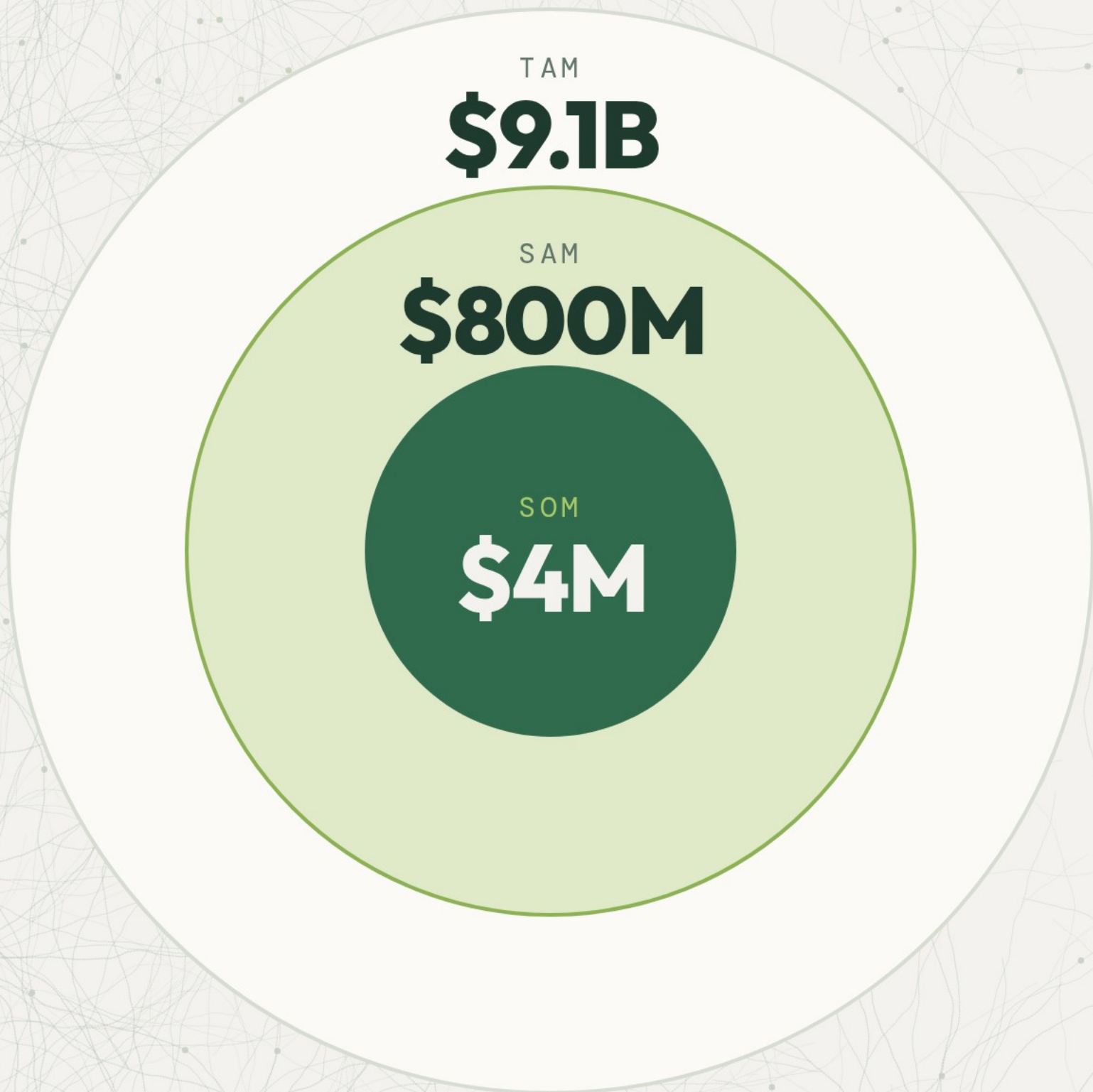
Stop competing on volume. Compete on fit.



Scores are our team's estimates, to be checked in customer interviews.

MARKET SIZE

From \$9.1B to our first \$4M.



TAM
\$9.1B

US & Canada protective packaging, 2025

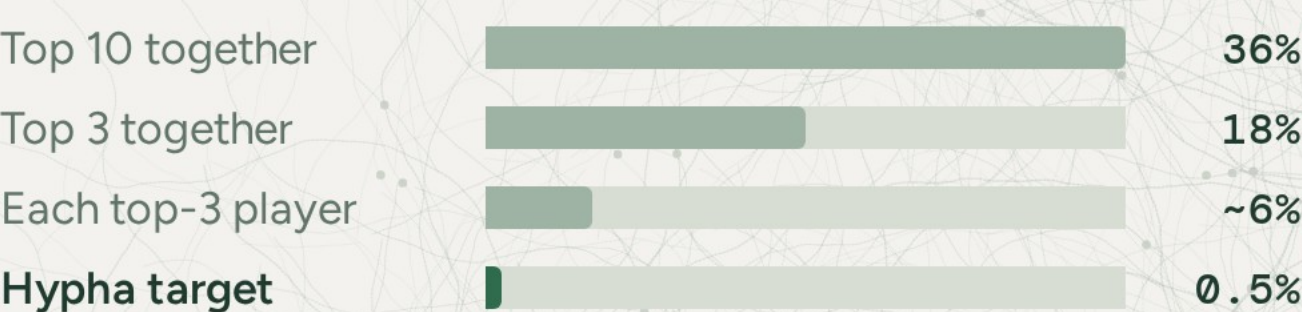
SAM
\$800M

E-commerce void fill

$$\begin{matrix} \$1.9\text{B void fill} \\ \times \\ 42.1\% \text{ e-commerce} \\ = \\ \$800\text{M} \end{matrix}$$

SOM
\$4M

0.5% of SAM, benchmarked against today's players



Conservative case: 0.25% gives about \$2M.

Circles not to scale. Sources: [1] Future Market Insights, US & Canada Protective Packaging Market · [2] Market Intelo, Void Fill Packaging Market (e-commerce share) · [3] Future Market Insights, Void Fill Packaging Systems Market Share Analysis. SOM share is our assumption.

BUSINESS MODEL

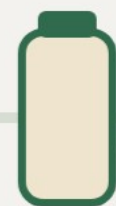
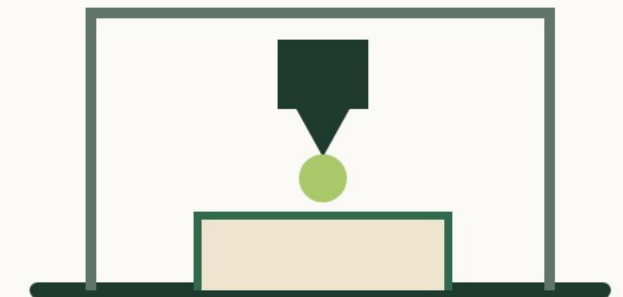
Lease the station. Sell the cartridges.



Station lease

Predictable monthly revenue

Your pack line



Material cartridges

Refilled for every box the station fills

Like a printer and its ink: revenue grows with every box shipped.

— HYPHA IN ONE SENTENCE

Hypha is a smart void-fill system that **3D-prints a biodegradable, mycelium-based foam** directly into the shipping box, expanding around each product in seconds for a **perfect custom fit**. No plastic, no waste, no guesswork.

Sources (1 of 2)

- [1]** Future Market Insights, Protective Packaging Industry Analysis in United States and Canada, 2025 to 2035 (TAM \$9.1B; void fill \$1.9B)
<https://www.futuremarketinsights.com/reports/united-states-and-canada-protective-packaging-market>
- [2]** Market Intelo, Void Fill Packaging Market Research Report 2034 (e-commerce = 42.1% of void fill, 2025)
<https://marketintelo.com/report/void-fill-packaging-market>
- [3]** Future Market Insights, Void Fill Packaging Systems Market Share Analysis (top 3 = 18%, top 10 = 36%)
<https://www.futuremarketinsights.com/reports/void-fill-packaging-systems-market-share-analysis>
- [4]** Market Research Future, Void Fill Packaging System Market (growth drivers)
<https://www.marketresearchfuture.com/reports/void-fill-packaging-system-market-38467>
- [5]** US EPA, Facts and Figures about Materials, Waste and Recycling: Plastics; Containers and Packaging
<https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/plastics-material-specific-data>
<https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/containers-and-packaging-product-specific>
- [6]** Our World in Data, "Packaging is the source of 40% of the planet's plastic waste"
<https://ourworldindata.org/data-insights/packaging-is-the-source-of-40-of-the-planets-plastic-waste>
- [7]** Beyond Plastics, The Real Truth About the U.S. Plastics Recycling Rate
<https://www.beyondplastics.org/publications/us-plastics-recycling-rate>
- [8]** The Recycling Partnership, State of Recycling: The Present and Future of Residential Recycling in the U.S. (2024)
https://recyclingpartnership.org/wp-content/uploads/dlm_uploads/2024/01/Recycling-Partnership-State-of-Recycling-Report-1.12.24.pdf

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- [9]** Smithers, "Packaging waste outpacing global capacity for sustainable management" (434.5M tonnes, 2025)
<https://www.smithers.com/resources/2025/december/packaging-waste-outpacing-global-capacity>
- [10]** Packaging Technology Today, "The Rise of Sustainable Packaging in the U.S.: Trends, Drivers, and the Road Ahead"
<https://www.packagingtechtoday.com/materials/the-rise-of-sustainable-packaging-in-the-u-s-trends-drivers-and-the-road-ahead/>
- [11]** Adobe Express, "The power of the package" consumer survey (73% say premium packaging improves view of brand)
<https://www.adobe.com/express/learn/blog/power-of-the-package>
- [12]** Packaging Technology Today, "Study Finds E-commerce Packaging Choices Impact Purchasing Decisions" (Ryder study; boxes chosen for protection)
<https://www.packagingtechtoday.com/featureds/study-finds-e-commerce-packaging-choices-impact-purchasing-decisions/>
- [13]** 2025 consumer study of 1,193 shoppers (unboxing and repeat purchase)
Cited by the team; link not yet added.
- [14]** Journal of Consumer Psychology, unboxing experiments
Cited by the team; link not yet added.
- [15]** Customer interview: Holiday (DTC apparel brand) team, conducted by Ledger, Group 11. Primary research.
Interview notes held by the team.

Notes: Calculated figures (21% void fill share, \$800M SAM, \$4M SOM) are derived from [1] to [3]. The SOM capture rate (0.5%) is a team assumption benchmarked against incumbent shares. Value-map scores and the additive-manufacturing scrap illustration are team estimates.